

MIDDLETON ISLAND, AK, USA

WMO: 703430

Lat: 59.433N

Lon: 146.333W

Elev: 100

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25402

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	20.8	24.6	7.2	8.0	23.8	9.8	9.1	26.3	48.5	36.4	44.1	37.5	15.2	20	0.798

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	63.4	57.3	61.2	56.3	59.0	55.3	58.8	62.5	57.4	60.9	56.1	59.0	8.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	69.3	60.1	55.3	65.3	58.4	54.6	63.6	57.9	25.5	62.3	24.7	60.9	23.9	58.8	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.7	35.9	31.6	19.2	65.9	4.3	3.5	16.2	68.4	13.6	70.4	11.2	72.4	8.1	74.9		
(5)			16.2	59.7	4.0	2.9	13.3	61.9	10.9	63.6	8.7	65.2	5.7	67.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	36.1	35.9	35.9	40.5	44.8	50.6	54.7	55.6	52.1	45.2	39.3	37.0		
	DBStd	8.32	5.24	4.81	4.67	3.33	4.48	2.98	2.78	2.59	3.09	3.82	4.96	4.41		
	HDD50	2623	430	394	436	285	166	25	0	13	151	321	402			
	HDD65	7652	895	814	901	734	625	433	319	292	388	613	771	867		
	CDD50	446	0	0	0	1	6	42	146	173	75	3	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	13.7	16.9	16.2	15.1	14.1	12.3	10.2	9.1	9.7	12.3	14.7	16.4	17.4		
	PrecAvg	76.8	7.4	6.2	6.5	5.5	4.7	3.3	4.4	6.2	8.5	9.3	7.5	7.3		
(15) Precipitation	PrecMax	94.0	15.1	10.6	22.3	9.2	9.7	5.0	6.5	12.6	13.8	16.4	17.3	15.1		
	PrecMin	47.3	1.2	2.4	3.5	2.8	1.6	1.4	1.7	1.5	2.9	3.9	2.4	3.0		
	PrecStd	9.2	2.9	2.2	3.3	1.9	2.0	0.9	1.0	2.3	2.6	3.3	3.2	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	45.4	45.9	53.7	57.4	63.4	65.9	68.0	62.8	54.4	48.5	46.2		
		MCWB	42.8	42.9	42.3	47.1	50.2	56.4	59.8	61.2	55.2	51.0	46.4	43.6		
	2%	DB	44.7	43.5	44.6	48.0	54.7	59.3	63.2	63.8	59.4	52.4	47.7	45.2		
		MCWB	42.1	40.6	41.3	43.4	49.3	54.2	57.7	58.2	54.6	49.5	46.0	43.0		
	5%	DB	43.3	43.0	43.2	46.0	52.3	56.8	61.0	61.2	57.3	51.7	46.2	44.4		
		MCWB	40.7	40.3	40.1	42.0	47.8	52.4	56.8	56.8	53.3	49.0	43.9	41.9		
	10%	DB	42.8	41.3	42.0	45.1	49.9	54.9	58.6	59.0	55.5	49.9	45.4	43.1		
		MCWB	40.3	39.1	39.2	41.2	46.5	51.4	55.3	55.8	52.4	47.6	42.8	40.3		
	0.4%	WB	44.5	43.2	44.0	47.8	51.8	57.8	60.4	62.2	57.2	52.3	48.0	46.0		
		MCDB	45.2	44.5	45.2	53.4	56.8	62.4	64.5	66.9	60.1	53.1	48.8	47.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.8	41.4	42.2	44.1	50.2	55.0	58.5	59.5	55.7	50.4	46.3	44.2		
		MCDB	44.4	43.3	44.2	46.7	54.3	59.1	62.3	63.1	58.4	51.7	46.8	45.2		
	5%	WB	41.5	40.4	40.7	42.5	48.3	53.2	57.3	57.9	54.6	49.0	45.6	42.8		
		MCDB	43.4	42.5	43.1	45.4	51.8	55.7	61.0	61.2	56.9	50.5	46.2	44.5		
	10%	WB	40.4	39.2	39.3	41.4	46.6	51.8	56.0	56.4	53.6	47.7	44.0	41.1		
		MCDB	42.6	41.5	41.8	44.6	49.0	54.7	58.8	59.2	55.6	49.1	45.6	43.4		
	5% DB	MDBR	4.7	4.7	5.7	6.5	6.4	6.3	5.8	5.6	5.7	4.9	4.8	4.9		
		MCDBR	4.9	5.0	6.5	8.6	10.1	9.0	8.9	8.5	7.5	5.8	4.7	5.5		
		MCWBR	5.3	5.0	5.5	6.4	6.4	6.0	6.2	5.9	5.7	4.9	5.0	5.3		
		MCWBR	5.1	4.9	6.0	8.4	9.5	8.1	8.7	8.4	6.7	5.0	4.4	5.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.4	5.0	5.6	6.1	6.3	5.9	6.1	5.9	5.6	5.0	5.1	5.6		
	Clear-Sky Solar Irradiance	taub	0.280	0.313	0.327	0.367	0.373	0.368	0.371	0.368	0.352	0.329	0.287	N/A		
		taud	2.257	2.390	2.429	2.321	2.372	2.451	2.481	2.489	2.485	2.383	2.223	N/A		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	167	225	259	264	271	275	271	263	248	213	157	N/A		
		Edh at Noon	13	20	25	34	34	32	31	29	24	20	13	N/A		
(44) All-Sky Solar Radiation	RadStd		89	272	690	1193	1524	1621	1408	1087	781	369	127	56		
			13	50	81	118	141	131	187	152	111	34	20	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page